

Probability and Hypothesis Testing Problems

Problem 1

A quality control engineer tests a batch of 100 electronic components. The probability that any single component is defective is 0.02, and components are independent.

- a) What is the probability that exactly 3 components are defective?
- b) What is the probability that at least 2 components are defective?
- c) What is the expected number of defective components?

Solution:

This is a binomial distribution problem with $n = 100$ and $p = 0.02$.

a) $P(X = 3) = C(100, 3) \times (0.02)^3 \times (0.98)^{97}$

Using the binomial formula: $P(X = 3) = 100! / (3! \times 97!) \times (0.02)^3 \times (0.98)^{97}$

Calculation yields approximately 0.1823

b) $P(X \geq 2) = 1 - P(X = 0) - P(X = 1)$

$P(X = 0) = (0.98)^{100} \approx 0.1326$

$P(X = 1) = 100 \times 0.02 \times (0.98)^{99} \approx 0.2707$

$P(X \geq 2) = 1 - 0.1326 - 0.2707 = 0.5967$

c) $E[X] = np = 100 \times 0.02 = 2$ defective components

Problem 2

A researcher wants to test the hypothesis that the mean height of a population is 68 inches. They take a sample of 50 individuals and find a sample mean of 69.2 inches with a sample standard deviation of 3.5 inches.

- a) State the null and alternative hypotheses.
- b) Calculate the test statistic.
- c) Determine the p-value for a two-tailed test at $\alpha = 0.05$.
- d) What is your conclusion?

Solution:

a) $H_0: \mu = 68$ inches

$H_1: \mu \neq 68$ inches

b) Since $n = 50 > 30$, we can use the z-test:

$z = (\bar{x} - \mu_0) / (s / \sqrt{n}) = (69.2 - 68) / (3.5 / \sqrt{50})$

$$z = 1.2 / (3.5/7.071) = 1.2 / 0.495 = 2.424$$

c) For a two-tailed test with $z = 2.424$:

$$P\text{-value} = 2 \times P(Z > 2.424) = 2 \times 0.0077 = 0.0154$$

d) Since $p\text{-value} (0.0154) < \alpha (0.05)$, we reject the null hypothesis. There is sufficient evidence to conclude that the mean height is different from 68 inches.

Problem 3

The following data represents test scores from two different teaching methods:

Method A	Method B
85	78
92	82
88	75
90	80
87	79

Conduct a two-sample t-test to determine if there is a significant difference between the two teaching methods. Use $\alpha = 0.05$.

Solution:

Method A: $n_1 = 5$, $\bar{x}_1 = 88.4$, $s_1 = 2.7$

Method B: $n_2 = 5$, $\bar{x}_2 = 78.8$, $s_2 = 2.6$

Pooled standard deviation: $s_p = \sqrt{((n_1-1)s_1^2 + (n_2-1)s_2^2)/(n_1+n_2-2)}$

$$s_p = \sqrt{[(4 \times 7.29 + 4 \times 6.76)/8]} = \sqrt{[56.1/8]} = 2.65$$

$$t = (\bar{x}_1 - \bar{x}_2) / [s_p \sqrt{(1/n_1 + 1/n_2)}]$$

$$t = (88.4 - 78.8) / [2.65 \sqrt{(1/5 + 1/5)}]$$

$$t = 9.6 / [2.65 \times 0.632] = 9.6 / 1.675 = 5.73$$

Degrees of freedom: $df = n_1 + n_2 - 2 = 8$

Critical value for two-tailed test at $\alpha = 0.05$: $t_{0.025,8} = 2.306$

Since $|t| = 5.73 > 2.306$, we reject H_0 . There is a significant difference between the two teaching methods.